

Review Packet #4 Kinetics and Thermo

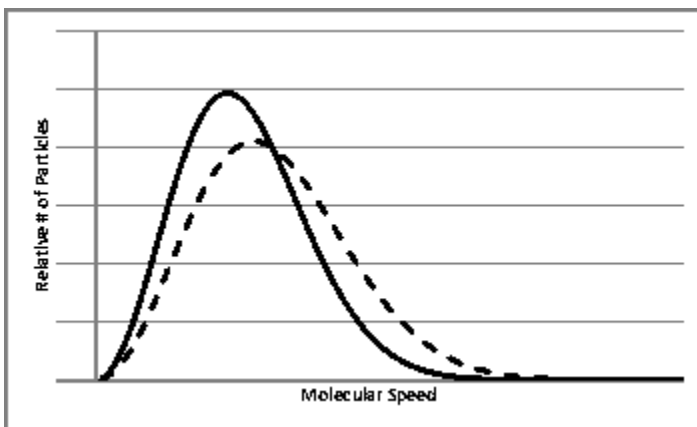
Multiple Choice

Identify the choice that best completes the statement or answers the question.

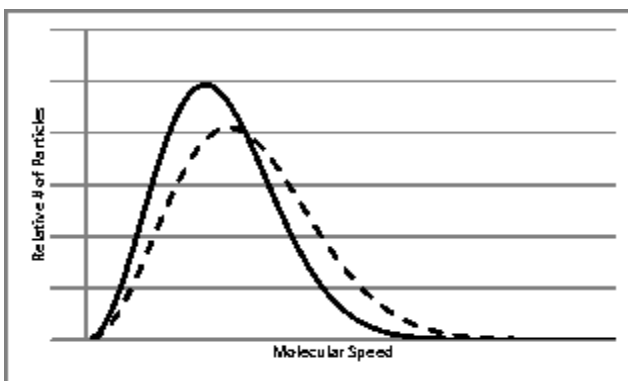
- _____ 1. The energy required to form the transition state in a chemical reaction
- Free energy
 - Activation energy
 - Ionization energy
 - Kinetic energy
 - Lattice energy
- _____ 2. Each of the following factors can affect the forward rate of a chemical reaction EXCEPT
- temperature
 - concentration of reactants of the forward reaction
 - presence of a catalyst
 - removal of some of the products of the forward reaction
 - physical state or state of subdivision of solid reactants
- _____ 3. The half-life for radioactive element X is 10.0 min. What weight of X was originally present in a sample if 40. grams is left after 60. minutes?
320. grams
 640. grams
 - 1,280 grams
 - 2400 grams
 - 2560 grams
- _____ 4. Which of the following has the least effect on the rate of a reaction?
- adding a solid catalyst to a gas phase reaction
 - adding a solid catalyst to a liquid phase reaction
 - adding inert miscible liquid to a liquid phase reaction
 - adding inert gas to a gas phase reaction at constant volume
 - adding excess base to a neutralization reaction for a nearly insoluble weak acid
- _____ 5. Refer to the steps of a mechanism proposed for the reaction of nitrogen monoxide with hydrogen.
- $\text{NO(g)} + \text{NO(g)} \leftrightarrow \text{N}_2\text{O}_2\text{(g)}$ fast equilibrium
- $\text{N}_2\text{O}_2\text{(g)} + \text{H}_2\text{(g)} \leftrightarrow \text{N}_2\text{O(g)} + \text{H}_2\text{O(g)}$ slow
- $\text{N}_2\text{O(g)} + \text{H}_2\text{(g)} \leftrightarrow \text{N}_2\text{(g)} + \text{H}_2\text{O(g)}$ fast
- Which rate law is consistent with this mechanism?
- $\text{rate} = k[\text{NO}]^2$
 - $\text{rate} = k[\text{NO}]^2[\text{H}_2]$
 - $\text{rate} = k[\text{N}_2\text{O}][\text{H}_2]$
 - $\text{rate} = k[\text{N}_2\text{O}_2][\text{H}_2]$
 - $\text{rate} = k[\text{N}_2\text{O}_2][\text{H}_2][\text{NO}]^2$

- _____ 6. $2A(g) + B(g) \rightleftharpoons 2C(g)$
When the concentration of substance B in the reaction above is doubled, all other factors being held constant, it is found that the rate of the reaction remains unchanged. The most probable explanation for this observation is that
- the order of the reaction with respect to substance B is 1
 - substance B is not involved in any of the steps in the mechanism of the reaction
 - substance B is not involved in the rate determining step of the mechanism, but is involved in subsequent steps
 - substance B is probably a catalyst, and as such, its effect on the rate of the reaction does not depend on its concentration
 - the reactant with the smallest coefficient in the balanced equation generally has little or no effect on the rate of the reaction
- _____ 7. Assume that $2A + B \rightarrow C$ is the rate determining step of a mechanism. 3.0 moles of A and 2.0 moles of B are placed in a 1.0 L flask; after 5 minutes the concentration of C reaches 1.0 M. After five minutes the rate will have
- increased by a factor of 9.
 - increased by a factor of 18.
 - decreased by a factor of 9.
 - decreased by a factor of 10
 - decreased by a factor of 18.
- _____ 8. For a given reaction at a temperature of 27 °C, the rate law is $\text{Rate} = k[X][Y]$. If the concentration of X and of Y are both 0.40 M, the rate is $4.0 \times 10^{-6} \text{ mol / L min}$. Determine the value of k (the rate constant) at this temperature.
- $2.5 \times 10^{-5} \text{ L/(mol min)}$
 - $2.5 \times 10^{-3} \text{ L/(mol min)}$
 - $2.5 \times 10^{-5} \text{ mol/(L min)}$
 - $1.0 \times 10^{-6} \text{ L/(mol min)}$
 - $1.0 \times 10^{-5} \text{ L/(mol min)}$
- _____ 9. The major reason an increase in temperature causes an increase in reaction rate is that
- the activation energy changes with temperature.
 - the fraction of high energy molecules increases.
 - the reactant molecule pressure increases.
 - molecules collide with greater frequency.
 - catalysts become more effective.
- _____ 10. Catalysts effectively increase reaction rate by
- increasing the K_{eq} .
 - increasing the concentration of the reactant.
 - decreasing the concentration of the products.
 - lowering the activation energy requirements.
 - decreasing the reaction temperature.
- _____ 11. It is found that in a certain first order reaction the half-life is 1.4 minutes. The rate constant, k, for that same temperature is about
- 0.35 min^{-1}
 - 0.50 min^{-1}
 - 0.71 min^{-1}
 - 2.0 min
 - 2.0 min^{-1}

12. The activation energy for a reaction is +30 kJ/mol; therefore the activation energy for the reverse reaction must be
- 30 kJ/mol
 - greater than 30 kJ/mol
 - less than 10 kJ/mol
 - +30 kJ/mol
 - a value that cannot be calculated without more data
13. The rate law expression is $\text{rate} = k[X]^2[Y]$ for a certain reaction. Both X and Y are reactant gases. If the volume of the system is reduced to 1/3 of the original volume, the relative new reaction rate will be
- 3 times greater
 - 9 times greater
 - 27 times greater
 - 27 times less
 - 9 times less



14. The speeds of carbon dioxide and nitrogen molecules were measured at 300 K in a 1.0 L flask. The graph above displays these results. The dashed line can be assumed to represent
- carbon dioxide because its maximum number of particles is less than that of nitrogen.
 - nitrogen because its maximum number of particles is less than that of carbon dioxide.
 - carbon dioxide because more of its molecules are moving at faster speeds due to its greater molecular weight.
 - nitrogen because more of its molecules are moving at faster speeds due to its lower molecular weight.



15. The speeds of nitrogen molecules were measured at 300 K and at 500 K in a 1.0 L flask. The graph above displays these results. The solid line can be assumed to represent the data at
- 300 K because at lower temperatures, a greater number of molecules move at slower speeds.
 - 300 K because at lower temperatures, the molecules move at greater speeds.
 - 500 K because at higher temperatures, the maximum number of particles is greater.
 - 500 K because at higher temperatures, the molecules move at lower speeds.
16. Which of the following is probably true for a solid solute with a highly endothermic heat of solution when dissolved in water?
- The solid has a low lattice energy.
 - As the solute dissolves, the temperature of the solution increases.
 - The resulting solution is ideal.
 - The solid is more soluble at higher temperatures.
 - The solid has a high energy of hydration.
17. The energy in a chemical or physical change that is available to do useful work
- activation energy
 - free energy
 - ionization energy
 - kinetic energy
 - lattice energy
18. The energy change that occurs in the conversion of an ionic solid to widely separated gaseous ions
- activation energy
 - free energy
 - ionization energy
 - kinetic energy
 - lattice energy
19. Which is a correct comparison of the characteristics of a catalyzed reaction to the corresponding characteristics of the same reaction without a catalyst present?
- Their energies of activation are the same.
 - Their enthalpies of reaction are the same.
 - Their free energies of reaction are the same.
- I only
 - II only
 - II and III only
 - I and III only
 - I, II, and III

- _____ 20. $\text{C}_2\text{H}_4(\text{g}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$
For the reaction of ethylene represented above, ΔH is -1323 kJ/mol. What is the value of ΔH if the combustion produced liquid water $\text{H}_2\text{O}(\text{l})$ rather than water vapor $\text{H}_2\text{O}(\text{g})$? (ΔH for the phase change $\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$ is -44 kJ/mol.)
- 1235 kJ/mol
 - 1279 kJ/mol
 - 1323 kJ/mol
 - 1367 kJ/mol
 - 1411 kJ/mol
- _____ 21. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
The reaction indicated above is thermodynamically favorable at 298 K, but becomes unfavorable at higher temperatures. Which of the following is true at 298 K?
- ΔG , ΔH , and ΔS are all positive.
 - ΔG , ΔH , and ΔS are all negative.
 - ΔG , and ΔH are negative, but ΔS is positive.
 - ΔG and ΔS are negative, but ΔH is positive.
- _____ 22. A student observes that an open container of water left overnight in the laboratory experienced a good deal of evaporation. Which of the following sets of conditions best describes the thermodynamics of this evaporation event?
- $\Delta H > 0$, $\Delta S > 0$
 - $\Delta H > 0$, $\Delta S < 0$
 - $\Delta H < 0$, $\Delta S > 0$
 - $\Delta H < 0$, $\Delta S < 0$
- _____ 23. Which gives the thermodynamic parameters for the phase change in a system that begins as an open container of liquid water placed in a constant temperature environment of 230 K?
- $\Delta G < 0$
 - $\Delta S < 0$
 - $\Delta G < 0$ and $\Delta H < 0$
 - $\Delta G < 0$, $\Delta H < 0$, and $\Delta S < 0$
- _____ 24. Which describes the direction of changes in enthalpy and entropy for an endothermic dissolving process of an ionic solute in water at constant temperature?
- Enthalpy increases and entropy decreases.
 - Enthalpy decreases and entropy increases.
 - Both enthalpy and entropy decrease.
 - Both enthalpy and entropy increase.
- _____ 25. When excess HCl reacts with 0.250 mole of NH_3 , a total of 11.0 kJ of heat energy is released. What is the value of ΔH in kJ/mol NH_3 consumed?
- 44.0
 - 11.0
 - 2.75
 - +11.0
 - +44.0

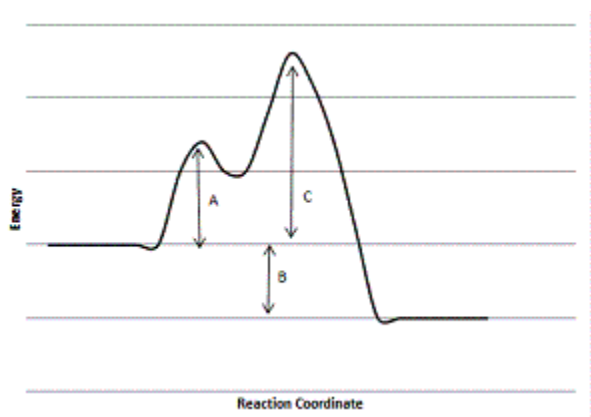
26. Which of the following reactions has the largest positive value of ΔS per mole of Cl_2 ?

- a. $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$
- b. $\text{Cl}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{Cl}_2\text{O}(\text{g})$
- c. $\text{Mg}(\text{s}) + \text{Cl}_2(\text{g}) \rightarrow \text{MgCl}_2(\text{s})$
- d. $2\text{NH}_4\text{Cl}(\text{s}) \rightarrow \text{N}_2(\text{g}) + 4\text{H}_2(\text{g}) + \text{Cl}_2(\text{g})$
- e. $\text{Cl}_2(\text{g}) \rightarrow 2\text{Cl}(\text{g})$

27. $3\text{C}_2\text{H}_2(\text{g}) \rightarrow \text{C}_6\text{H}_6(\text{g})$

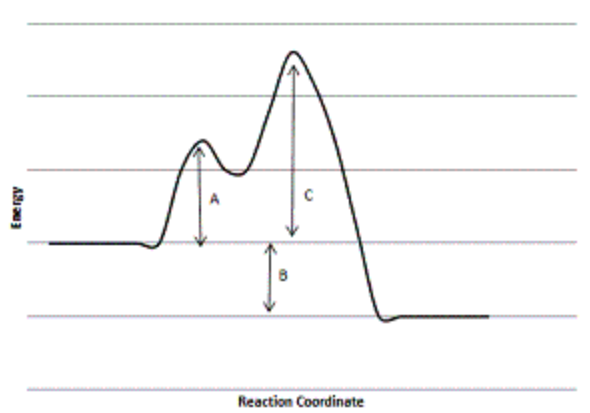
What is the standard enthalpy change, ΔH_r , for the reaction represented above? (ΔH_f of $\text{C}_2\text{H}_2(\text{g})$ is 230 kJ/mol; ΔH_f of $\text{C}_6\text{H}_6(\text{g})$ is 83 kJ/mol)

- a. -607 kJ/mol
- b. -147 kJ/mol
- c. -19 kJ/mol
- d. +19 kJ/mol
- e. +773 kJ/mol



28. Use the potential energy diagram shown above to answer the questions below. The activation energy of the rate determining step is given by

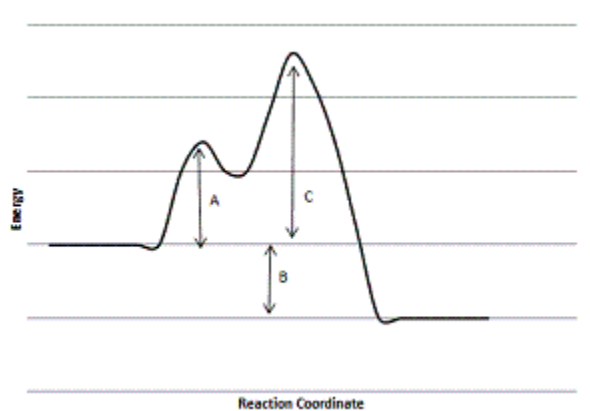
- a. A
- b. B
- c. C
- d. A + C



29.

Use the potential energy diagram shown above to answer the questions below. The activation energy of the 1st step in the mechanism of the reverse reaction is given by

- A
- B
- C
- $B + C$

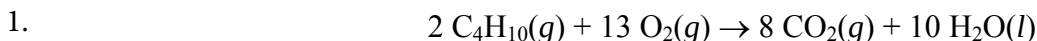


30.

Use the potential energy diagram shown above to answer the questions below. The enthalpy change of the reaction is given by

- A
- B
- C
- $A + C$

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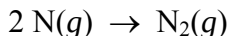


The reaction represented above is spontaneous at 25°C. Assume that all reactants and products are in their standard state.

- Predict the sign of ΔS° for the reaction and justify your prediction.
- What is the sign of ΔG° for the reaction? How would the sign and magnitude of ΔG° be affected by an increase in temperature to 50°C? Explain your answer.
- What must be the sign of ΔH° for the reaction at 25°C? How does the total bond energy of the reactants compare to that of the products?
- When the reactants are placed together in a container, no change is observed even though the reaction is known to be spontaneous. Explain this observation.

2. Answer the following questions that relate to the chemistry of nitrogen.

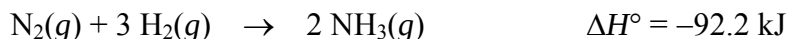
- Two nitrogen atoms combine to form a nitrogen molecule, as represented by the following equation.



Using the table of average bond energies below, determine the enthalpy change, ΔH , for the reaction.

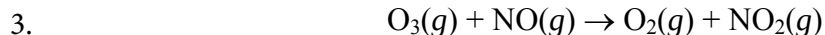
Bond	Average Bond Energy (kJ mol ⁻¹)
N – N	160
N = N	420
N ≡ N	950

- The reaction between nitrogen and hydrogen to form ammonia is represented below.



Predict the sign of the standard entropy change, ΔS° , for the reaction. Justify your answer.

- The value of ΔG° for the reaction represented in part (b) is negative at low temperatures but positive at high temperatures. Explain.
- When $\text{N}_2(g)$ and $\text{H}_2(g)$ are placed in a sealed container at a low temperature, no measurable amount of $\text{NH}_3(g)$ is produced. Explain.



Consider the reaction represented above.

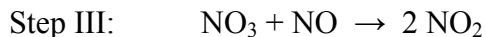
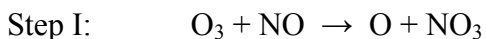
- (a) Referring to the data in the table below, calculate the standard enthalpy change, ΔH° , for the reaction at 25°C . Be sure to show your work.

	$\text{O}_3(g)$	$\text{NO}(g)$	$\text{NO}_2(g)$
Standard enthalpy of formation, ΔH_f° , at 25°C (kJ mol^{-1})	143	90.	33

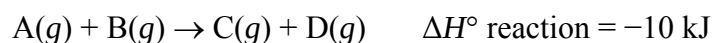
- (b) Make a qualitative prediction about the magnitude of the standard entropy change, ΔS° , for the reaction at 25°C . Justify your answer.
- (c) On the basis of your answers to parts (a) and (b), predict the sign of the standard free-energy change, ΔG° , for the reaction at 25°C . Explain your reasoning.
- (d) Use the information in the table below to write the rate-law expression for the reaction, and explain how you obtained your answer.

Experiment Number	Initial $[\text{O}_3]$ (mol L^{-1})	Initial $[\text{NO}]$ (mol L^{-1})	Initial Rate of Formation of NO_2 ($\text{mol L}^{-1} \text{s}^{-1}$)
1	0.0010	0.0010	x
2	0.0010	0.0020	$2x$
3	0.0020	0.0010	$2x$
4	0.0020	0.0020	$4x$

- (e) The following three-step mechanism is proposed for the reaction. Identify the step that must be the slowest in order for this mechanism to be consistent with the rate-law expression derived in part (d). Explain.



4. Consider the following general equation for a chemical reaction.



- (a) Describe the two factors that determine whether a collision between molecules of A and B results in a reaction.
- (b) How would a decrease in temperature affect the rate of the reaction shown above? Explain your answer.
- (c) Write the rate law expression that would result if the reaction proceeded by the mechanism shown below.



- (d) Explain why a catalyst increases the rate of a reaction but does not change the value of the equilibrium constant for that reaction.

Packet #4 Answers

Multiple Choice

- | | | | | |
|------|-------|-------|-------|-------|
| 1. B | 7. E | 13. C | 19. C | 25. A |
| 2. D | 8. A | 14. D | 20. D | 26. D |
| 3. E | 9. B | 15. A | 21. B | 27. A |
| 4. D | 10. D | 16. D | 22. A | 28. C |
| 5. B | 11. B | 17. B | 23. D | 29. D |
| 6. C | 12. E | 18. E | 24. D | 30. B |

Free Response

1.
 - a. $\Delta S < 0$. 15 moles of gas is producing 8 moles of gas and 10 moles of liquid. Therefore, disorder is decreasing.
 - b. $\Delta G < 0$. Given that the reaction is spontaneous, ΔG must be negative. As temperature increases, the value of ΔG becomes more positive. Since ΔS is negative, increasing temperature makes the $-T\Delta S$ term a larger positive. Thus, the magnitude of ΔG decreases as the reaction becomes less spontaneous.
 - c. ΔH must be negative for the reaction at 25°C . Since ΔS is negative, ΔH must be negative for the reaction to be spontaneous. Thus, the total bond energy of the reactants is less than that of the bond energy of the products.
 - d. The activation energy must be supplied in order for the reaction to begin. Without activation energy, no products will be formed.
2.
 - a. $\Delta H = -950 \text{ kJ/mol}$. This is calculated using bonds broken (which are none since two individual nitrogen atoms are forming a bond) minus bonds formed, which is the triple bond between nitrogen atoms.
 - b. $\Delta S < 0$ as 4 moles of gas produces 2 moles of gas showing a decrease in disorder.
 - c. Since ΔH and ΔS are both negative, at low temperatures, the value of ΔH is responsible for the spontaneity of the reaction, but at high temperatures, the value of ΔS is responsible for reaction spontaneity. $\Delta G = \Delta H - T\Delta S$. At higher temperatures, the $-T\Delta S$ term becomes more and more positive.
 - d. No activation energy is supplied for the reaction to occur.
3.
 - a. $\Delta H = \text{products} - \text{reactants}$; $\Delta H = -200 \text{ kJ/mol}$
 - b. The magnitude of ΔS will be small given that two moles of gas are producing two moles of gas. Therefore, there is not a large change in disorder.
 - c. Since ΔH is largely negative and ΔS has a very small magnitude, ΔG will most likely be negative as well as the $-T\Delta S$ term is mostly negligible.
 - d. $\text{Rate} = k[\text{O}_3]^1[\text{NO}]^1$; In experiments #1 and 2, while $[\text{O}_3]$ remained constant, $[\text{NO}]$ doubled as did the rate of formation of NO_2 . Therefore the reaction is first order with

respect to NO. In experiments #1 and 3, while [NO] remained constant, [O₃] doubled as did the rate of formation of NO₂. Therefore, the reaction is first order with respect to O₃.

- e. The first step is the rate determining (slowest) step as the rate law from this elementary step matches the experimental rate law.

4.

- a. One factor that determines whether a reaction occurs is the force/energy with which the two molecules collide. If they do not collide with enough energy to exceed the activation energy, no reaction will occur. Another factor is the orientation of the two molecules. If the orientation of the two molecules is not correct, the right parts of the molecules will not interact in a way that allows a reaction.
- b. The rate of reaction would decrease with a decrease in temperature. A temperature decrease slows the gas molecules down. Thus, the molecules do not hit each other as often or with as much force. Therefore, the rate of reaction slows down.
- c. $\text{Rate} = k[\text{A}][\text{B}]^2$
- d. A catalyst provides an alternate mechanism for the reaction to occur, thus lowering the activation energy. It does not affect the energy or amounts of the products or reactants present at equilibrium, thus K does not change.

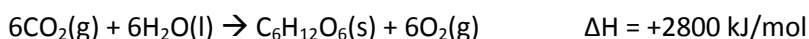
Multiple Choice - No calculator may be used.☐

1. In which of the following circumstances is the value for K_{eq} always greater than 1?

ΔH	ΔS
a. Positive	Positive
b. Positive	Negative
c. Negative	Negative
d. Negative	Positive

☐

2. Consider the following reaction showing photosynthesis:



Which of the following is true regarding the thermal energy of the system?

- a. It is transferred from the surroundings to the reaction.
- b. It is transferred from the reaction to the surroundings.
- c. It is transferred from the reactants to the products.
- d. It is transferred from the products to the reactants.

☐

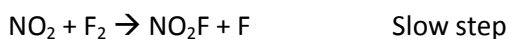
3. $\text{SO}_2\text{Cl}_2 \rightarrow \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$

At 600 K, SO_2Cl_2 will decompose to form sulfur dioxide and chlorine gas via the above equation. If the reaction is found to be first order overall, which of the following will cause an increase in the half-life of SO_2Cl_2 ?

- a. Increasing the initial concentration of SO_2Cl_2 .
- b. Increasing the temperature at which the reaction occurs.
- c. Decreasing the overall pressure in the container.
- d. None of these will increase the half-life.

☐

4. A proposed mechanism for a reaction is as follows:



What is the order of the overall reaction?

- | | |
|----------------|-----------------|
| a. Zero order | c. Second order |
| b. First order | d. Third order |

☐

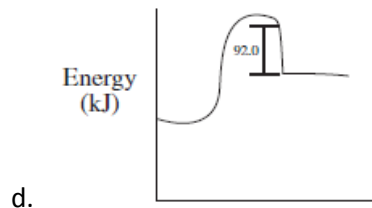
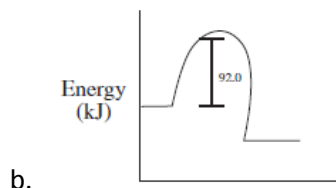
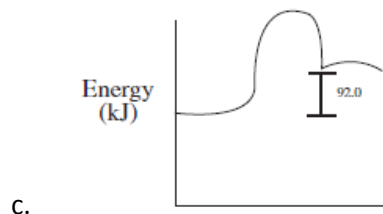
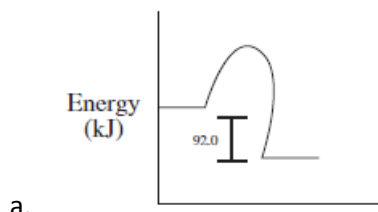
5. $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$

The above reaction will experience a rate increase by the addition of a catalyst such as platinum. Which of the following best explains why?

- a. The catalyst increases the overall frequency of collisions in the reactant molecules.
- b. The catalyst increases the frequency of collisions that occur at the proper orientation in the reactant molecules.
- c. The catalyst introduces a new reaction mechanism for the reaction.
- d. The catalyst increases the activation energy for the reaction.



Which of the following energy diagrams gives an accurate representation of the above reaction?



Questions 7-10 refer to the following.

Inside a calorimeter, 100.0 mL of 1.0 M hydrocyanic acid (HCN), a weak acid, and 100.0 mL of 0.50 M sodium hydroxide are mixed. The temperature of the mixture rises from 21.5°C to 28.5°C. The specific heat of the mixture is approximately 4.2 J/g °C, and the density is identical to that of water (1 g/mL).



7. Identify the correct net ionic equation for the reaction that takes place.

- $\text{HCN(aq)} + \text{OH}^-\text{(aq)} \leftrightarrow \text{CN}^-\text{(aq)} + \text{H}_2\text{O(l)}$
- $\text{HCN(aq)} + \text{NaOH(aq)} \leftrightarrow \text{NaCN(aq)} + \text{H}_2\text{O(l)}$
- $\text{H}^+\text{(aq)} + \text{OH}^-\text{(aq)} \leftrightarrow \text{H}_2\text{O(l)}$
- $\text{H}^+\text{(aq)} + \text{CN}^-\text{(aq)} + \text{Na}^+\text{(aq)} + \text{OH}^-\text{(aq)} \leftrightarrow \text{H}_2\text{O(l)} + \text{CN}^-\text{(aq)} + \text{Na}^+\text{(aq)}$



8. What is the approximate enthalpy change for the reaction?

- 1.5 kJ
- 2.9 kJ
- 5.9 kJ
- 11.8 kJ



9. As the temperature of the reaction mixture increases, what happens to the equilibrium constant and why?

- The equilibrium constant increases because more products are created.
- The equilibrium constant increases because the rate of the forward reaction increases.
- The equilibrium constant decreases because the equilibrium shifts to the left.
- The value for the equilibrium constant is unaffected by temperature and will not change.

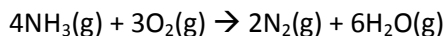


10. If the experiment is repeated with 200.0 mL of 1.0 M HCN and 100. mL of 0.50 M NaOH, what would happen to the values for ΔT and the ΔH ?

- | ΔT | ΔH_{rxn} |
|------------------|-------------------------|
| a. Increase | increase |
| b. Stay the same | stay the same |
| c. Decrease | stay the same |
| d. Stay the same | increase |

Free Response – A calculator may be used.

1. Ammonia gas can be burned in air via the following reaction:



The standard entropy of formation values for the varying substances are listed in the table below.

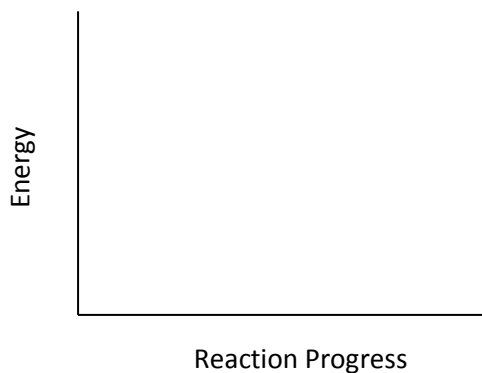
Substance	$S^\circ(\text{J/mol K})$
$\text{NH}_3(\text{g})$	193
$\text{O}_2(\text{g})$	205
$\text{N}_2(\text{g})$	192
$\text{H}_2\text{O}(\text{g})$	189

- a. Calculate the value for the change in entropy, ΔS , for the overall reaction.

Several bond enthalpies are listed in the table below.

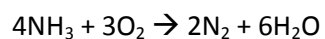
Bond	Enthalpy (kJ/mol)	Bond	Enthalpy (kJ/mol)
N – H	388	O=O	498
O – H	463	N≡N	941

- b. Calculate the enthalpy change, ΔH , for the overall reaction.
- c. Is this reaction thermodynamically favored at 25°C? Justify your answer.
- d. On the reaction coordinate below, draw a line showing the progression of this reaction. Label both ΔH and E_a on the graph.



Name _____

The following data were gathered during a rate experiment for the reaction of ammonia with oxygen:



Experiment	Initial $[\text{NH}_3]$ (mol L ⁻¹)	Initial $[\text{O}_2]$ (mol L ⁻¹)	Initial Rate of appearance of N_2 (mol L ⁻¹ min ⁻¹)
1	0.0100	0.0100	3.74×10^{-3}
2	0.0300	0.0100	3.37×10^{-2}
3	0.0100	0.0200	1.50×10^{-2}

e. In experiment #1, determine the initial rate of appearance of H_2O .

f. Determine the order of reaction with respect to

i. NH_3 . Show your reasoning.

ii. O_2 . Show your reasoning.

g. Write the rate law for the reaction.

h. Determine the value of the constant k . Specify units.

Packet #4 Quiz Solutions

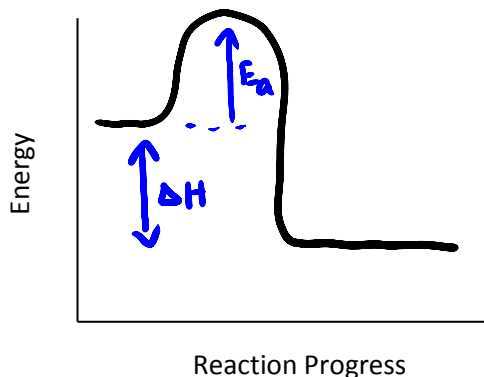
Multiple Choice

- | | |
|------|-------|
| 1. D | 6. C |
| 2. A | 7. A |
| 3. D | 8. C |
| 4. C | 9. C |
| 5. C | 10. C |

Free Response (12 pts total)

- a. $\Delta S = 131 \text{ J/mol K}$ [1 pt for correct calculation]
- b. $\Delta H = -1288 \text{ kJ/mol}$ [1 pt for set up with correct numbers of each bond; 1 pt for correct calculation]
- c. The reaction is thermodynamically favored because $\Delta G < 0$. [1 pt for answer and explanation; must either calculate ΔG or reference that $\Delta H < 0$ and $\Delta S > 0$]

d.



[1 pt for demonstration of an exothermic reaction; 1 pt for properly labeling both ΔH and E_a]

- e. $1.12 \times 10^{-2} \text{ M/min}$ [1 pt for correct calculation]
- f.
- In experiments #1 and #2, $[\text{O}_2]$ is constant while $[\text{NH}_3]$ triples as the rate of appearance multiples by 9. Therefore, the reaction is 2nd order with respect to NH_3 . [1 pt for correct order with justification – may be algebraic as well]
 - In experiments #1 and #3, $[\text{NH}_3]$ is constant while $[\text{O}_2]$ doubles as the rate of appearance quadruples. Therefore, the reaction is 2nd order with respect to O_2 . [1 pt for correct order with justification – may be algebraic as well]
- g. $\text{Rate} = k[\text{NH}_3]^2[\text{O}_2]^2$ [1 pt for rate law that is consistent with orders given in part f]
- h. $k = 3.74 \times 10^5 \text{ M}^{-3} \text{ min}^{-1}$ [1 pt for correct calculation of k from rate law in part g; 1 pt for correct units for k that are consistent with the rate law in part g]